Compression pipeline, device profile, and accuracy–efficiency tradeoff

Team: REPLACE_ME Submitted file: submission/student_model_fp32.onnx Baseline: student_mobilenetv2_distilled.pt

1. Compression pipeline

The starting network is the official student checkpoint: torchvision MobileNetV2 with the classifier replaced by Linear(1280, 10), 2.237 million parameters. ResNet-50 distillation is already stored in that checkpoint, so this pipeline does not repeat it. The graph the judges feed stays NCHW float32, input name input, shape (1, 3, 224, 224), ImageNet mean and standard deviation. The output name is output, ten logits, in torchvision EuroSAT order: AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, SeaLake.

Three transforms were available, and a candidate is kept only when it retains at least 97% of baseline accuracy and its worst class is at least as accurate as the baseline worst class. Export uses torch.onnx.export at opset 13 with a fixed batch of 1. Duplicate value_info names are stripped so a later AI Hub compile does not reject the graph.

Resolution. Native EuroSAT frames are 64×64 and are only stretched to 224×224. The file still accepts 224×224. A bilinear resize (align_corners=False) then maps that tensor to the working resolution before the MobileNetV2 stem. The search compared 224, 160, and 128. The submitted working size is 160.

Structured pruning. torch-pruning removes whole channels by global group-L2 magnitude, rounds widths to multiples of 8, and leaves the first convolution and the 10-way linear layer in place. Unstructured zeroing was skipped because it leaves parameter count, file size, and MAC count unchanged. A probe that removed about 25% of channels fell to roughly 13% accuracy before any recovery, so the submitted model keeps all 2.237 million parameters.

Static quantization. ONNX Runtime quantize_static writes QDQ int8, Entropy calibration with a MinMax fallback, per-channel QInt8 weights, and QUInt8 activations. Only Conv ops are quantized; the final 10-way layer stays fp32, using at least 128 training images as the calibration set. Dynamic quantization was left out because device compilers often reject it. After the resize, recovery is about 80 AdamW steps at learning rate 1e-4 with class-weighted cross-entropy (AnnualCrop 1.4, Highway 1.3, Pasture 2.0, PermanentCrop 1.8, River 2.2). The teacher file was absent, so the repair uses EuroSAT labels only.

2. Qualcomm AI Hub device profile

Chosen profile: Qualcomm Arduino UNO Q. The Software Track pre-competition packet names that board as the shared target so every team is compiled and profiled on one device. If the UNO Q is missing from the account catalog, the fallback in profile_hub.py is the closest low-power device that catalog lists. The rationale is the same for either board: the accuracy-safe saving on this 10-class land-use model is arithmetic, from 0.326 to 0.166 billion MACs per image, which is the cost that dominates on a small low-power part. The compile specification remains the contest shape, input (1, 3, 224, 224), with the 160-pixel resize inside the graph.

Compile and profile jobs were not submitted. No API token was configured, so this report claims no on-device latency, memory, or energy. Section 3 is the host-side tradeoff. After qai-hub configure, the submitted file can be profiled with profile_hub.py --model submission/student_model_fp32.onnx --device set to the UNO Q name returned by the account catalog.

3. Accuracy versus efficiency

Baseline and the submitted model were measured back-to-back in ONNX Runtime on CPU, after garbage collection, on the same machine. Ratios are baseline divided by the new value for cost, and new accuracy divided by baseline accuracy.

Metric	Baseline (224, fp32)	Submitted (160, fp32)	Change
Accuracy	99.19%	98.63%	99.4% retained
Worst class	95.85%	96.05% (PermanentCrop)	floor held
Parameters	2.237 million	2.237 million	1.00×
File size	8.51 MB	8.51 MB	1.00×
MACs / image	0.326 billion	0.166 billion	1.96× fewer
Latency	4.96 ms	2.57 ms	1.93× faster
Peak process RAM	471 MB	478 MB	0.98×

Energy / image	0.158 J	0.0959 J	1.65× less

Curve point, not submitted. The 160-pixel Conv-only int8 graph is 2.58 MB (3.30× smaller) at the same 0.166 billion MACs, with host latency 2.51 ms and estimated energy 0.119 J. Accuracy is 94.41% (95.2% of baseline) and the worst class is 84.09%. Both sit under the 97% retention rule and the worst-class floor, so the smaller file stays on the curve.

The submitted model therefore trades a 0.56 percentage-point drop in overall accuracy for 1.96× fewer MACs, 1.93× lower host latency, and 1.65× lower estimated energy, while the worst class moves from 95.85% to 96.05%. Parameter count and file size do not improve, so the part of the rubric that requires both a smaller parameter count and a smaller file is still unmet. An earlier full-validation pass at 128 pixels, unpruned fp32, reached 98.04% accuracy and 0.107 billion MACs, and made River the worst class at 95.17%, below the baseline worst class. That is why 160 is the working size in the hand-in. Latency from that earlier pass is omitted; only the paired rows above share a measurement session.

Conditions. Windows 10, Intel64 Family 6 Model 140 Stepping 1, GenuineIntel, 16 GB RAM, CPU only. Validation is a deterministic 80/20 split, seed 42, 5,400 images, because val_split.json was not in the materials used; judges should rescore on the official split. Energy is CodeCarbon in tdp_estimate mode: no power sensor was available, so joules are runtime times a typical wattage, labeled as an estimate, and computed the same way for baseline and final. Peak RAM is process resident set with the ONNX Runtime session loaded. It is not the weight footprint. Rebuild with: python run_pipeline.py --refine --team-name "TEAM".